Choreographies for Program Understanding

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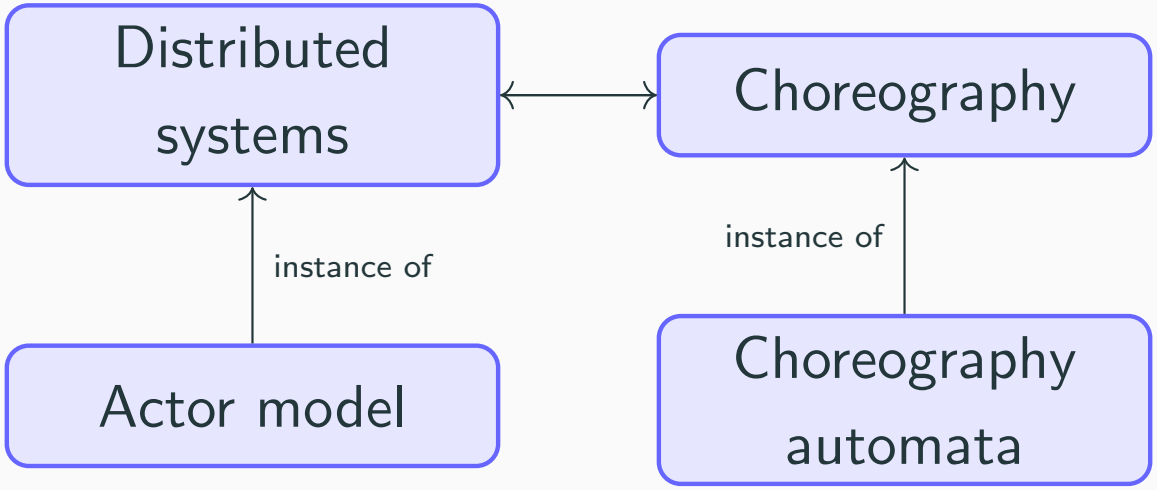
Distributed Systems are Hard

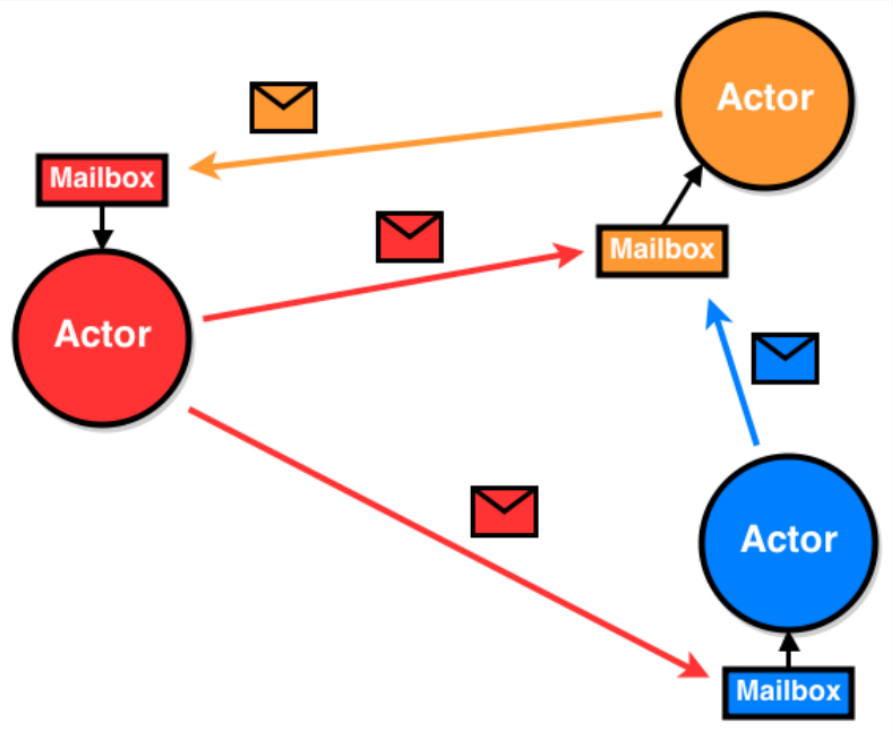
Abstractions in order to simplify:



The Actor Model

Main concepts

- Processes with mailbox
- Asynchronous messaging



The Actor Model

Ecosystem

- Erlang, Elixir, Scala
- Akka (Java), Actix (Rust)

Choreography Automata [1]

Informally

- Choreography: describes distributed protocols
- Paired with automata theory

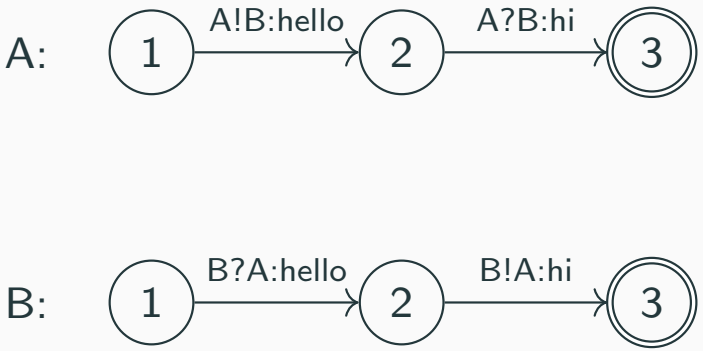
[1] Barbanera et al. "Choreography automata." COORDINATION, 2020.

Global vs Local View

Global

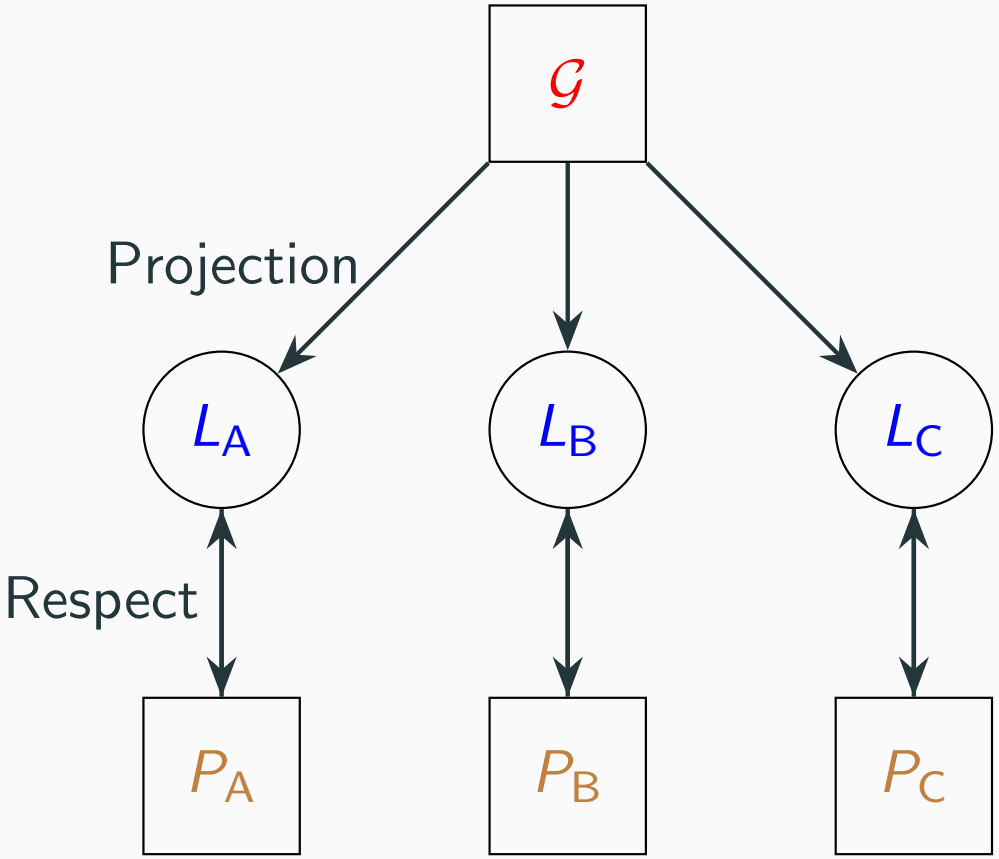
The communication system is seen as a whole.

Local



A participant's individual perspective.

Usual practice: Top-Down Approach



1. **Global view**
2. **Local views**
3. **Processes**

Usual practice: Top-Down Approach

Steps:

1. Write the global specification
2. Project to obtain the local specifications
3. Write local programs that respect the specifications

Problem: difficult to integrate existing code and architecture.

Aim

Problem:

Debug and understand legacy code

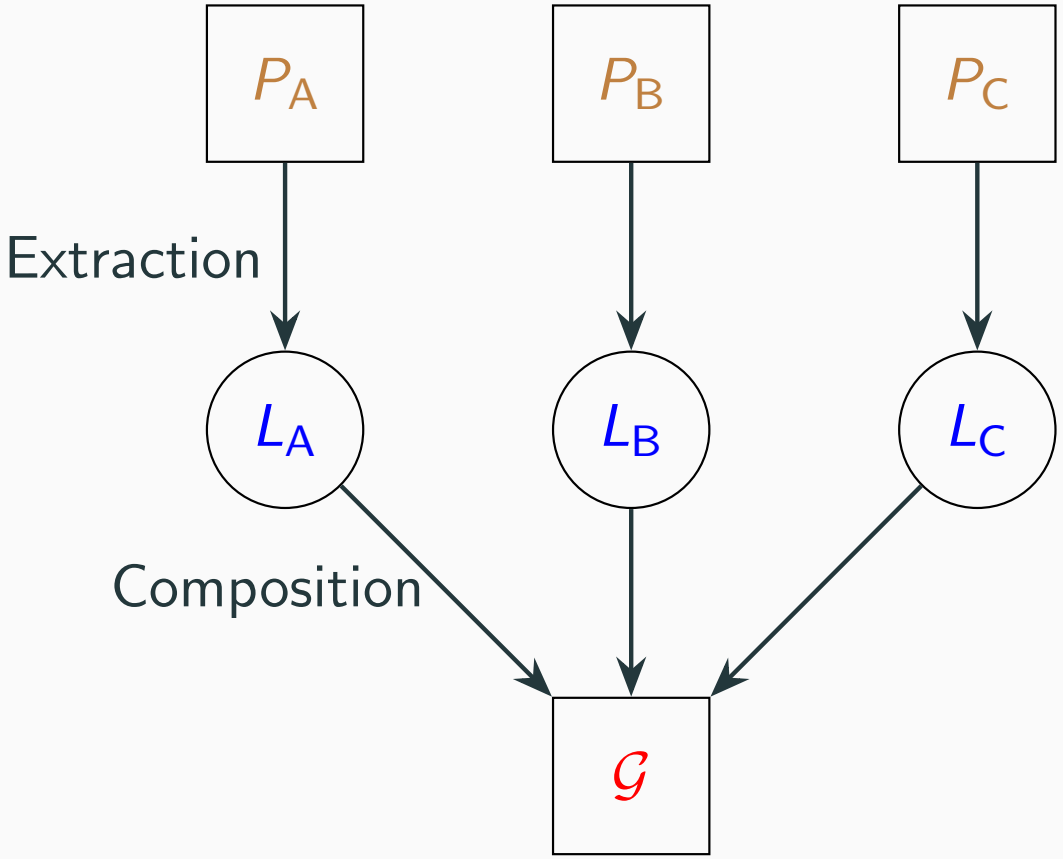
How?

Extract global specification from code

Support

Tool based on statical analysis

Bottom-up Approach



1. **Processes**
2. **Local view**
3. **Global view**

Bottom-up approach

Extraction steps

1. Analyze an input source code
2. Extract the local views
3. Compose local views to create an approximated global view

Output: an abstraction that captures all the possible behaviors.

Why

Possible benefits

- Improves understanding of the code
  - Give all good behavior
- Highlight bugs
  - Like deadlock, race condition, etc...

Requirements

- Automatic extraction
- Target mainstream languages
- Support creation and removal of participants
- Capture good behaviors and highlight misbehaviors
- Use a simple notation as output

Extraction steps

1. Take the file input and perform some simple analysis
2. Extract a local-view for each actor found
3. Create a global-view using the local-views
4. Minimize and analyse the output graph

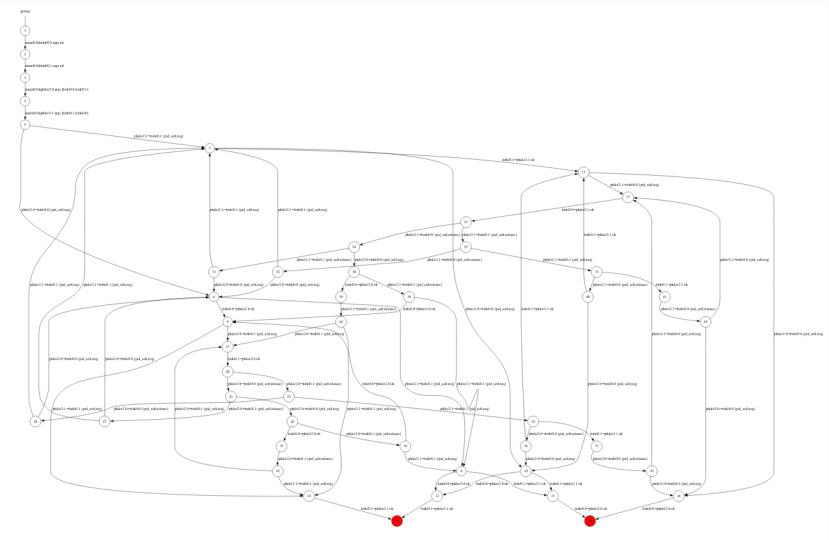
Problems - Undecidability

It would imply deciding termination. Choice of:

- Under-approximation
- Over-approximation

Problems - Huge descriptions

- Distributed system are naturally huge
- Needs of abstraction mechanisms: remove redundant information



Motivating Examples

Two examples:

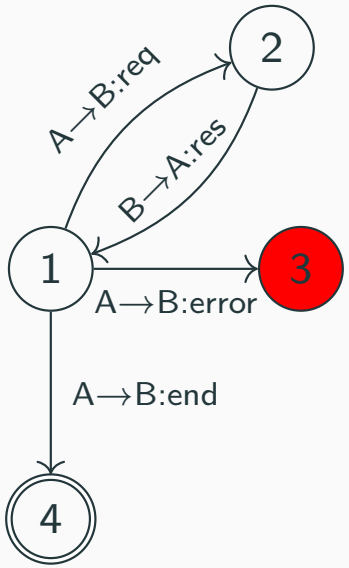
Dining Philosophers
(deadlock)

Bank Account
(race condition)

Deadlock

Final state: all the participants are in a *local* final state.

Deadlock: a non-final state without outgoing transitions.

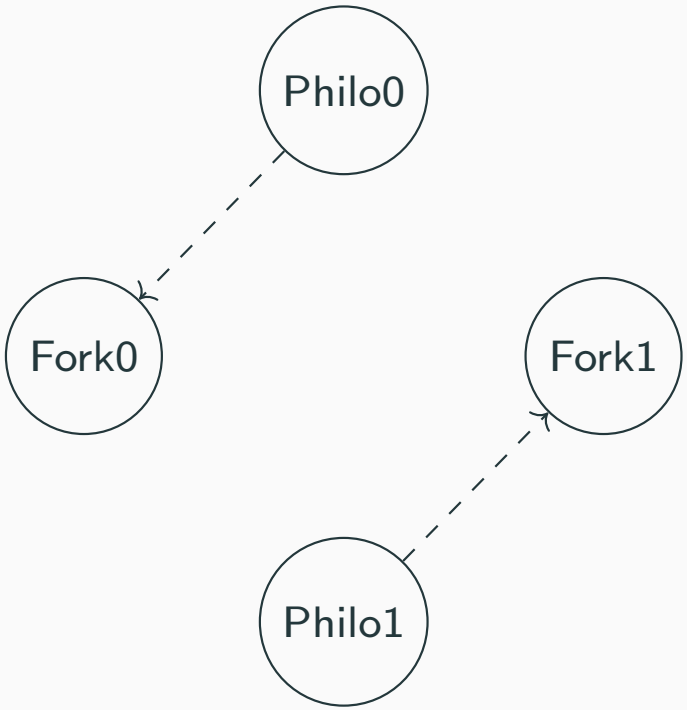


Dining Philosophers example

```
philosopher(Fork1, Fork2) →
  send req to Fork1,
  receive ack from Fork1,
  send req to Fork2,
  receive ack from Fork2,
  eat(),
  send release to Fork1,
  send release to Fork2,
  philosopher(Fork1, Fork2).

fork() →
  receive req from Phil,
  send ack to Phil,
  receive release from Phil,
  fork().
```

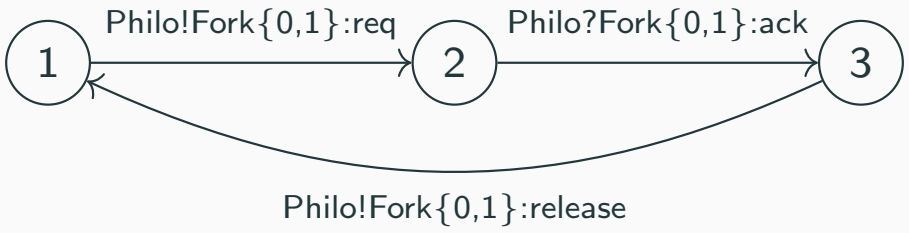
Participants:

- 2 philosophers
- 2 forks



Dining Philosophers example's local views

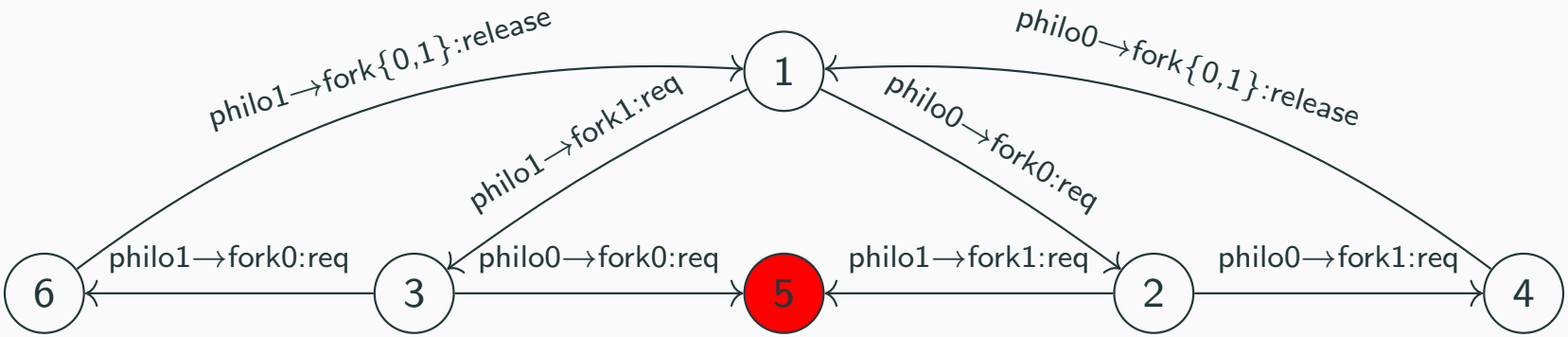
Note: merge of redundant communication!



Philosopher's local view.

Fork's local view.

Dining Philosophers global-view



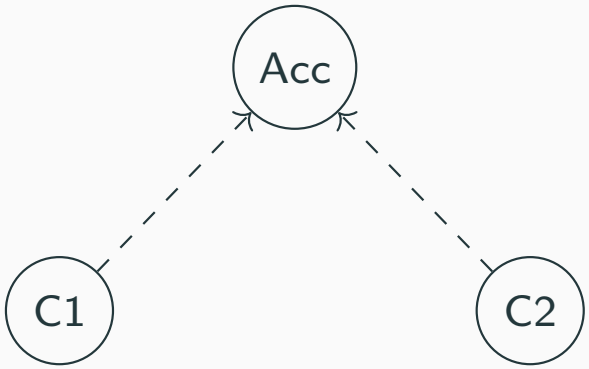
Global view of the Dining Philosophers example.

Bank Account example

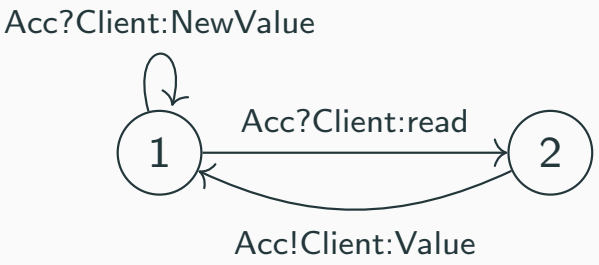
```
account(Value) →
  receive
    read from Client →
        send Value to Client,
        account(Value);
    NewValue from Client →
      account(NewValue).

client() →
  send read to Acc,
  receive Value from Acc,
  % operations on Value
  send NewValue to Acc.
```

Participants:

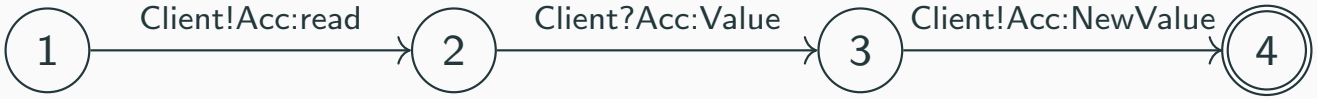
- 1 account
- 2 clients



Bank Account example's local views

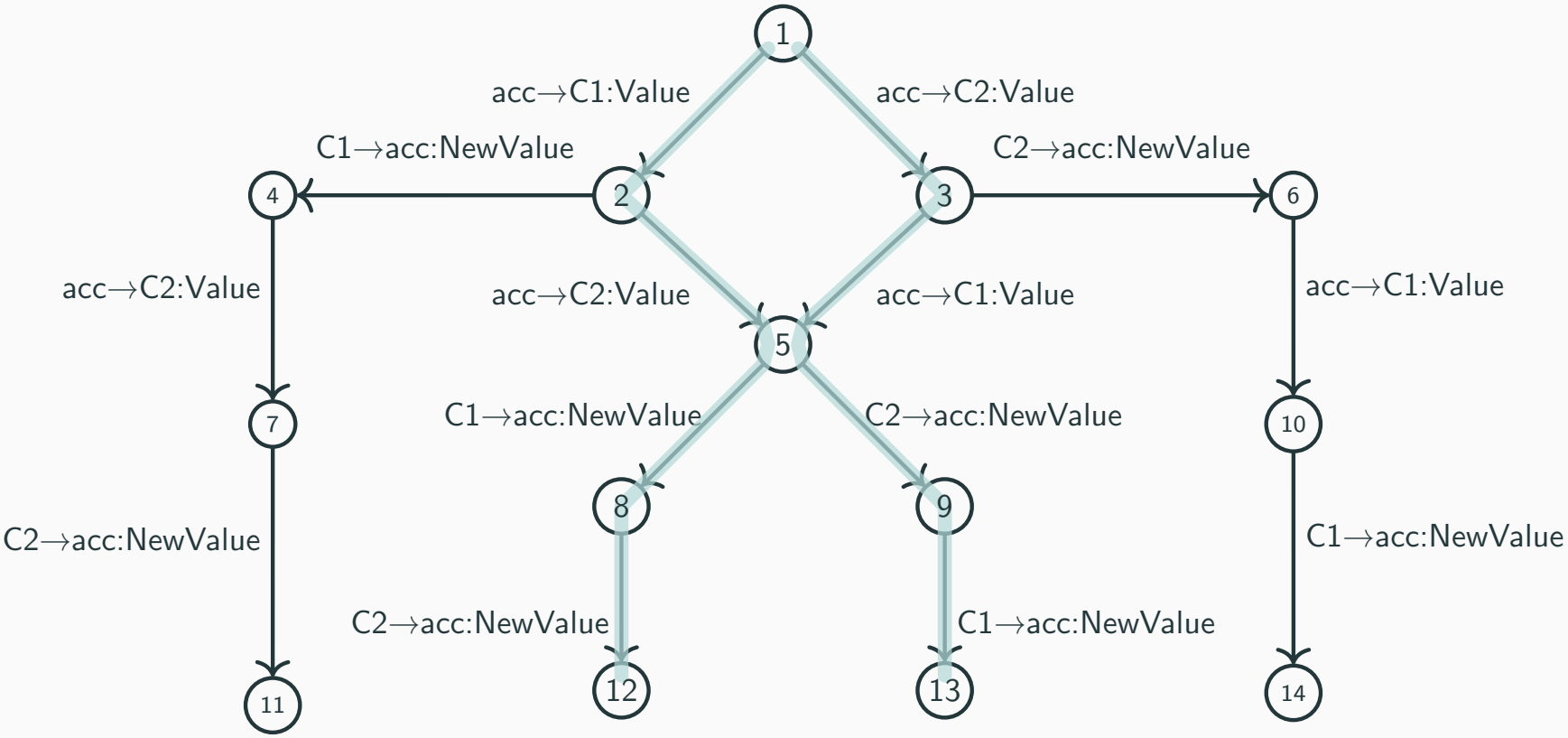


Account's local view.



Clients' local view.

Bank Account global-view



Global view of the Bank Account example.

Conclusion

Summary

- Problem: difficult to express legacy code
- Bridge the gap between programming and theoretical frameworks
- Use of bottom-up and over-approx techniques

Future work: create the tool as described (WIP on `github.com/gabrielegenovese/chorer`) and evaluate it on real-world examples.

Thanks for listening!